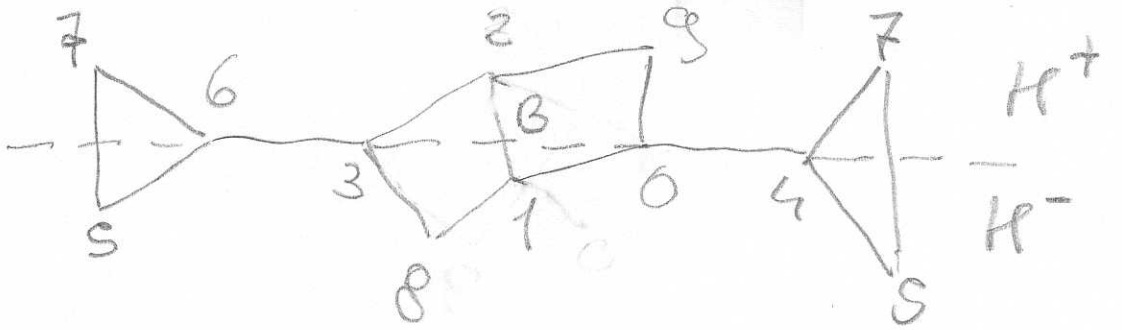
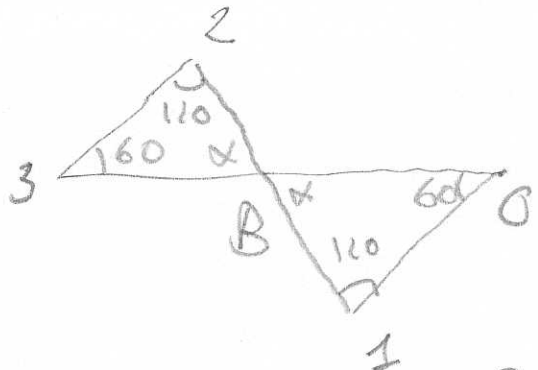
Explanation 601



The point 6, 3, 0 and 4 are all on the same equator γ, which bisects the angles in 6 and 4 of the triangles and in 0 and 3 of the quadrilaterals.

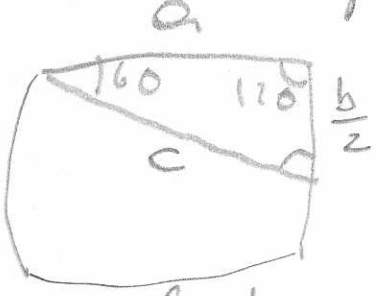
The equator γ cuts the sphere into two hemispheres H^+ and H^-. 2 must be in H^+ because 32 cuts in H^+, while 8 must be in H^-. Likewise 9 must be in H^+ (because 09 cuts H^+) and 1 in H^-.

So γ intersects the edge 12 in a point B



So the triangles 32B and B1C are congruent.

But it is not possible for the bisector of the angle of a quadrilateral to split the opposite side into half:



In fact we have the equations

$$\begin{cases} \tan\frac{a}{2}\tan\frac{b}{2}=\frac{1}{2} & (1) \\ \frac{b}{2}+c=\pi & (2) \\ \cos c=\cos a\cos\frac{b}{2}-\frac{1}{2}\sin a\sin\frac{b}{2} & (3) \end{cases}$$

Substituting (2) in (3) we get

$$-\cos\frac{b}{2} = \cos a \cos\frac{b}{2} - \frac{1}{2}\sin a \sin\frac{b}{2}$$

and (because $\pi > b$ and so $\cos\frac{b}{2} > 0$)

$$-1 = \cos a - \frac{1}{2}\sin a \tan\frac{b}{2} \qquad (4)$$

Substituting (1) in (4)

$$-1 = \cos a - \frac{1}{4}\sin a \cos\frac{a}{2} / \sin\frac{a}{2}$$

$$-1 = \cos^2\frac{a}{2} - \sin^2\frac{a}{2} - \frac{1}{2}\cos^2\frac{a}{2}$$

$$-1 = \frac{1}{2}\cos^2\frac{a}{2} - \sin^2\frac{a}{2}$$

But $\cos\frac{a}{2} > 0$ and so $-\sin^2\frac{a}{2} > -1$